Dissecting the Aggregate Market Elasticity

Victor Duarte
UIUC

Mahyar Kargar
UIUC

Dejanir Silva
Purdue

January 4, 2026

Why is the aggregate stock market so price-inelastic?

- Stock prices are very sensitive to flows
  - if there is a 1% flow (as a frac. of equity values) from bonds to stocks
  - the equity market value goes up by 5%
  - **inelastic market hypothesis** (Gabaix-Koijen, 2020)
- Can frictionless models generate a low aggregate elasticity?
- If not, what frictions help explain large price movements?
- Understanding this helps us answer a key open puzzle:
  - the origins of **high stock market volatility**

This paper

- Provide a framework to answer these questions
- Quantitative GE model with rich investor heterog. & financial frictions
- The **general equilibrium** perspective is key for **macro** elasticity
  - most existing work is PE and/or with limited/no heterogeneity
- Using perturbation methods to obtain analytical solution
- We consider two types of frictions
  - passive investments by households
  - heterog. active investors who face leverage constraints

Results (so far)

- Volatility depends on **aggregate elasticity** and **flow shocks**
- Aggregate elasticity:
  - is state-dependent: wealth distribution matters
  - no price impact of flows in frictionless settings (infinite elasticity)
  - adding passive investors makes market more inelastic
  - preference heterog. attenuates the price impact (risk misallocation)
  - binding leverage constraints amplifies the price impact
- Decompose the price impact into components due to:
  - risk premium
  - risk-free rate
  - price drift

Literature

- Aggregate (macro) demand elasticity:

  "change in agg. stock market's value in response to a \$1 flow to stocks from bonds"

  - Johnson (2006), Deuskar-Johnson (2011), Gabaix-Koijen (2021), Haddad-Huebner-Loualiche (2021), Li-Pearson-Zhang (2021)

- Micro demand elasticity:

  "change in relative price of stocks A and B if one buys \$1 of A, and sells \$1 of B"

  - Shleifer (1986), Wurgler-Zhuravskaya (2002), Chang-Hong-Liskovich (2014), Koijen-Yogo (2019)
  - higher elasticity than the aggregate market

- Heterogeneous agent models with financial frictions

  - Basak-Cuoco (1998), Gârleanu-Pedersen (2011), Longstaff-Wang (2012), Gârleanu-Panageas (2015), Silva (2020), Kargar (2021)

Model

Assets and endowments

Endowment: continuous time, infinite horizon, endowment economy

$$\frac{dY_t}{Y_t} = \mu dt + \sigma dZ_t$$

Markets: agents have access to two assets

- a riskless asset in zero net supply
- a risky claim on the aggregate endowment in unit supply

Asset return:

$$dR_t = \frac{dP_t + Y_t dt}{P_t} \equiv \mu_{R,t} dt + \sigma_{R,t}\, dZ_t,$$

- μ_R and σ_R determined in equilibrium

Investors

- $J+1$ types: $j = 0, 1, \ldots, J$ with mass ω_j
- Agents die at rate κ; mass $\kappa\omega_j$ born every period (to make the model stationary)
- All agents have Duffie-Epstein recursive preferences
  - risk aversions γ_j and common EIS ψ
- Choose consumption $C_{j,t}$ and portfolio weight in risky asset $\alpha_{j,t}$
- Two investor categories: Active & Passive

Investors (contd.)

- Passive investors:
  - type $j = 0$; have portfolio share $\alpha_{0,t} = \overline{\alpha}_{p,t}$ follow an exog. CIR process:

$$d\overline{\alpha}_{p,t} = \theta_p \left(\overline{\alpha} - \overline{\alpha}_{p,t}\right) dt + \sigma_p \sqrt{\overline{\alpha}_{p,t}}\, dZ_t$$

  - captures fluctuations in the passive investors risky asset holding
  - innovations to $\overline{\alpha}_p$: **portfolio flow shocks**
  - movements in $\overline{\alpha}_p$: due to agg. and/or indep. pure flow shocks

- Active investors:
  - type $j = 1, \ldots, J$; continuously rebalance their portfolios
  - face state-dependent **leverage constraints**

$$\alpha_{j,t} \leq \frac{\overline{\sigma}}{\|\sigma_{R,t}\|}$$

Investors' problem

Investor type j's problem

$$V_{j,t} = \max_{\{C_j, \alpha_j\}} \mathbb{E}_t \left[\int_t^\infty f_j \left(C_{j,s}, V_{j,s} \right) ds \right],$$

s.t. budget constraint

$$\frac{dW_{j,t}}{W_{j,t}} = \left[r_t + \left(\mu_{R,t} - r_t \right) \alpha_{j,t} - \frac{C_{j,t}}{W_{j,t}} \right] dt + \alpha_{j,t} \, \sigma_{R,t} \, dZ_t,$$

and $\alpha_{j,t} \in \Omega_{j,t}$ where

$$\Omega_{0,t} = \left\{ \alpha_0 : \alpha_0 = \overline{\alpha}_{p,t} \right\},$$

$$\Omega_{j,t} = \left\{ \alpha_j : \alpha_j \leq \frac{\overline{\sigma}}{\|\sigma_{R,t}\|} \right\}, \quad \text{for} \quad j = 1, \ldots, J.$$

Equilibrium Characterization

Investors' problem

- With homothetic preferences, value function for investor j:

$$V_{j,t} = \left(\frac{\xi_{j,t}}{\rho^{\psi}}\right)^{\frac{1-\gamma_j}{1-\psi}} \frac{W_{j,t}^{1-\gamma_j}}{1-\gamma_j}$$

- Consumption-wealth ratio: $\dfrac{C_{j,t}}{W_{j,t}} = \xi_{j,t}$.

- Convenient to consider investor j's **risk exposure** σ_j:

$$\sigma_{j,t} \equiv \alpha_j \|\sigma_R\| = \min\left\{\frac{\eta_t}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi-1}\frac{\sigma_{\xi_j,t}\sigma'_{R,t}}{\|\sigma_{R,t}\|}, \overline{\sigma}\right\}$$

  - $\eta_t \equiv \dfrac{\mu_{R,t} - r_t}{\|\sigma_{R,t}\|}$ is the Sharpe ratio of the risky asset

Aggregate state variables

- Wealth share of investor j

$$x_{j,t} \equiv \frac{\omega_j W_{j,t}}{P_t}$$

- Aggregate state variable

$$X_t = (x_t, \overline{\alpha}_{p,t}),$$

where $x_t \equiv (x_{1,t}, x_{2,t}, \ldots, x_{J,t})$

Connection between volatility and flows

- Volatility of the price-dividend ratio

$$\sigma_p\left(x, \overline{\alpha}_p\right) = \frac{p_x}{p}\sigma_x + \underbrace{\frac{p_{\overline{\alpha}_p}}{p}}_{\varepsilon_M^{-1}}\sigma_{\overline{\alpha}_p}$$

- First term: usual dependence on the wealth distribution
- Volatility also depends on the market elasticity & flow shocks
  - Need **both** inelastic markets and high flow shocks
- To compute ε_M, need to solve
  - system of high-dimensional PDEs with several state variables

State-global Perturbation

A family of economies

- Consider a family of economies indexed by $\epsilon > 0$.
- ϵ controls the degree of preference heterogeneity

$$\gamma_j = \gamma\left(1 + \hat{\gamma}_j \epsilon\right), \quad \text{where} \quad \sum_{j=0}^{J} \omega_j \hat{\gamma}_j = 0$$

  - γ: pop.-weighted avg risk aversion; $\hat{\gamma}_j$: prop. deviation from γ
- Risky asset demand by passive investors is also indexed by ϵ:

$$\overline{\alpha}_p = 1 + \hat{\alpha}_p \epsilon$$

- Finally, ϵ controls the tightness of the leverage constraint:

$$\overline{\sigma} = \|\sigma\| + \hat{\sigma}\epsilon$$

State-global perturbations

- Endogenous variables depend on agg. state variable X_t and ϵ
  - For instance, we have: $y(X, \epsilon)$ and $\xi_j(X, \epsilon)$
- We are interested in a **second-order** expansion, e.g.,

$$y(X, \epsilon) = y_0(X) + y_1(X)\epsilon + y_2(X)\epsilon^2 + \mathcal{O}(\epsilon^3)$$

  - k-th order corrections $y_k(X)$, $k \in \{0, 1, 2\}$: functions of state variable X
- Frictions are assumed to be 1st-order, so
  - 2nd-order allows us to capture the **interaction** btw. key frictions

State-global perturbations (contd.)

- Different from standard perturbation of DSGE models
  - no need to assume the aggregate state is near steady-state
  - requires solving for arbitrary functions of X instead of coefficients
  - provides a **global** method with respect to the states
- Previously used in
  - partial equilibrium: Fouque-Papanicolaou-Sircar (2000), McQuade (2018)
  - perturbation of pref. parameters: Kogan-Uppal (2001), Hansen-Heaton-Li (2008)
  - an application of the state-global perturbation: Kargar-Passadore-Silva (2020)
  - accuracy? convergence?

The benchmark economy: Constant expected returns

Lemma (Benchmark economy)

Suppose $\rho > (1-\psi^{-1})\left(\mu - \frac{\gamma\sigma^2}{2}\right)$. *Then, for the* $\epsilon = 0$ *economy,*

(i) *Investors' consumption-wealth ratio and risk exposure are given by:*

$$\xi_{j,0}(X) = \rho - (1-\psi^{-1})\left(\mu - \frac{\gamma\sigma^2}{2}\right), \quad \sigma_{j,0}(X) = \sigma, \quad \text{for } j = 0, 1, \ldots, J.$$

(ii) *Sharpe ratio, risk-free rate, and dividend yield are given by:*

$$\eta_0(X) = \gamma\sigma, \quad r_0(X) = \rho + \psi^{-1}\mu - (1+\psi^{-1})\frac{\gamma\sigma^2}{2}, \quad y_0(X) = \xi_{j,0}(X).$$

- The benchmark economy behaves like the frictionless Lucas (1978)

First-order terms

Proposition (First-order correction)

(i) *consumption-wealth ratio and risk exposures*

$$\xi_{j,1}(X) = (1-\psi)\gamma\sigma^2\left[\frac{r_1(X)}{\gamma\sigma^2} + \frac{\eta_1(X)}{\gamma\sigma} - \frac{\hat{\gamma}_j}{2}\right], \quad \sigma_{j,1}(X) = \min\left\{\frac{\eta_1(X)}{\gamma} - \frac{\eta_0(X)}{\gamma}\hat{\gamma}_j, \hat{\sigma}\right\}$$

(ii) *Sharpe ratio, risk-free rate, and dividend yield*

$$\eta_1(X) = \gamma\sigma\left[\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j - \frac{\hat{\alpha}_p x_0 + \frac{\hat{\sigma}}{\sigma}x_c}{1-x_0-x_c}\right], \quad y_1(X) = \left(1-\psi^{-1}\right)\gamma\sigma^2\sum_{j=0}^{J}x_{j,t}\frac{\hat{\gamma}_j}{2}$$

$$r_1(X) = \sigma^2\left[-\frac{\eta_1(X)}{\gamma\sigma} + \left(1-\psi^{-1}\right)\sum_{j=0}^{J}x_{j,t}\frac{\hat{\gamma}_j}{2}\right]$$

where $x_{u,t} \equiv \sum_{j\in\mathcal{J}_t^u} x_{j,t}$ *and* $x_{c,t} \equiv \sum_{j\in\mathcal{J}_t^c} x_{j,t}$.

Second-order terms

Simplification for today

- 3 agents $j = 0, 1, 2$: one passive, two actives
- 2 key state variables:
  1. wealth share of active investors:

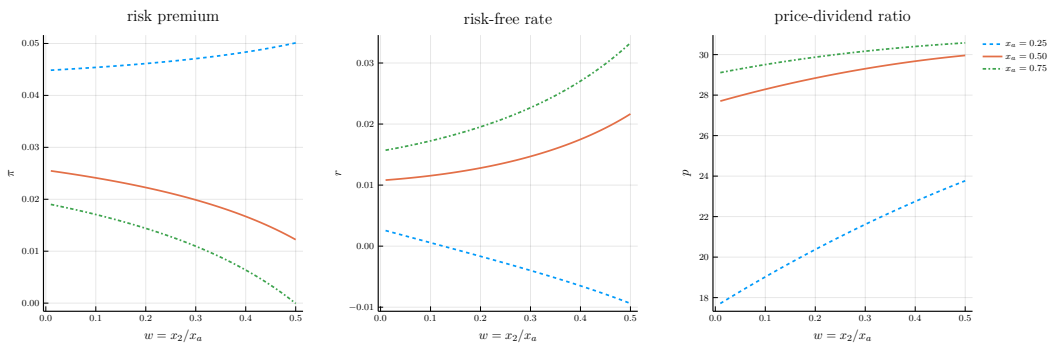
$$x_a = 1 - x_0$$

  2. wealth share of the least risk averse among active investors:

$$w = \frac{x_2}{x_a}$$

- No time-variation in passive portfolio share:
  - constant $\overline{\alpha}_p$ ($\theta_p = \sigma_p = 0$)
  - focusing on **permanent** changes from flows

Risk premium, risk-free rate, price-dividend ratio



- as active investors become undercapitalized (low x_a): $\pi \uparrow$, $r \downarrow$, $p \downarrow$

Aggregate Market Elasticity

Aggregate elasticity

- Consider passive flow F from riskless to the risky asset

$$F(X, \epsilon) \equiv \frac{(1 + \hat{\alpha}_p \epsilon)\, W_0 - W_0}{P_0} = \hat{\alpha}_p \epsilon\, x_0$$

  - normalized by the value of equities (following Gabaix-Koijen, 2021)
  - corresponds to passive investors
    - permanently changing their stock holdings from W_0 to $W_0(1 + \hat{\alpha}_p\, \epsilon)$
    - financed by selling riskless bonds

Aggregate elasticity (contd.)

- Define the inverse aggregate market elasticity as

$$\varepsilon_M^{-1} \equiv \frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)},$$

  - where $p = 1/y$ is the price-dividend ratio

- Second-order expansion of the price-dividend ratio:

$$p(X,\epsilon) = p_0(X) + p_1(X)\epsilon + p_2(X)\epsilon^2$$

Flows have no price impact up to first-order!

Proposition (First-order impact of portfolio flows)

(i) *The first-order impact of portfolio flows on the price-dividend ratio:*

$$\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = 0 + \mathcal{O}(\epsilon).$$

(ii) *The first-order impact of portfolio flows on the risk premium and the risk-free rate:*

$$\frac{\partial \pi(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{\gamma\sigma^2}{x_a} + \mathcal{O}(\epsilon), \qquad \frac{\partial r(X,\epsilon)}{\partial F(X,\epsilon)} = \frac{\gamma\sigma^2}{x_a} + \mathcal{O}(\epsilon).$$

Need to go beyond first-order

$$\frac{\partial \pi(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{\gamma\sigma^2}{x_a} + \mathcal{O}(\epsilon), \qquad \frac{\partial r(X,\epsilon)}{\partial F(X,\epsilon)} = \frac{\gamma\sigma^2}{x_a} + \mathcal{O}(\epsilon).$$

- Up to first-order
  - the risk premium and interest rate effects exactly offset each other
  - flows do not impact the price $\Rightarrow \varepsilon_M = \infty$
- Interpretation
  - frictions are 1st-order $\Rightarrow$ interaction of flows & frictions is 2nd-order
  - effectively we have a frictionless economy up to first-order
  - different previous work (Johnson, 2006; Gabaix-Koijen, 2021)
    - both investor heterogeneity & GE effects matter

Market Elasticity: Passive demand

Proposition (Aggregate elasticity: passive demand)

If there is no preference heterogeneity and active investors do not face leverage constraints, the inverse aggregate market elasticity, $1/\varepsilon_M$, is given by:

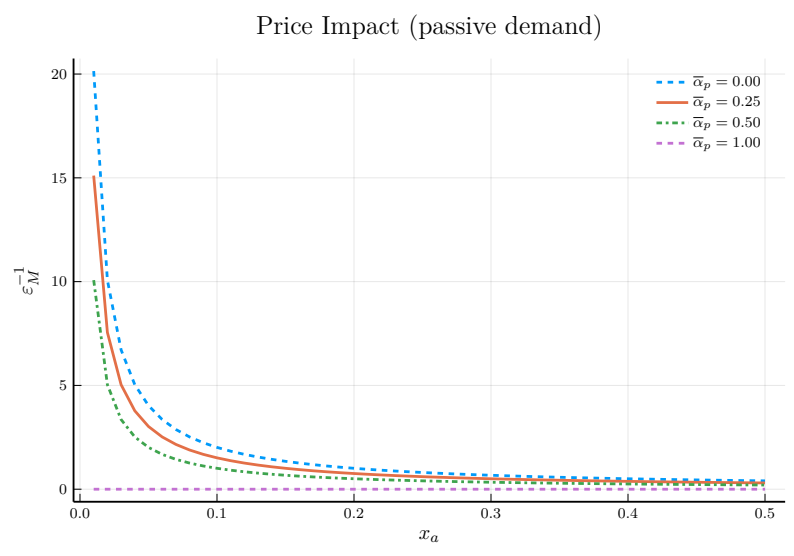
$$\varepsilon_M^{-1} = \left(1 - \psi^{-1}\right) \frac{\gamma\sigma^2}{y_0(X)} \frac{1 - \overline{\alpha}_p}{x_a} + \mathcal{O}\left(\epsilon^2\right).$$

Determinant of the market elasticity: Passive demand

$$\varepsilon_M^{-1} = \left(1 - \psi^{-1}\right) \frac{\gamma\sigma^2}{y_0(X)} \frac{1 - \overline{\alpha}_p}{x_a} + \mathcal{O}\left(\epsilon^2\right)$$

- If start at full passive participation $\overline{\alpha}_p = 1$ (effectively Lucas, 1978):
  - infinite elasticity as before (no price impact)
- If no passive market participation $\overline{\alpha}_p = 0$ (e.g., Basak-Couco, 1998):
  - flows have larger price impact (active risk concentration)
- Role of GE effects: sign of ε_M depends on EIS, ψ (absent in Gabaix-Koijen, 2021)
  - $\psi > 1$: risk premium effect dominates risk-free rate effect
- ε_M is time-varying & state-dependent
  - lower elasticity when active investors are undercapitalized (low x_a)

Inverse market elasticity: Passive demand



Investor heterogeneity

Proposition (Aggregate elasticity: preference heterogeneity)

If active investors have heterogeneous risk aversions but face no leverage constraints, the inverse aggregate market elasticity, $1/\varepsilon_M$, is given by:

$$\varepsilon_M^{-1} = \left(1 - \psi^{-1}\right) \frac{\gamma\sigma^2}{y_0(X)} \left[\frac{1 - \overline{\alpha}_p}{x_a} - \frac{\gamma_0 - \mathbb{E}^u\left[\gamma_j\right]}{\gamma}\right] + \mathcal{O}\left(\epsilon^2\right),$$

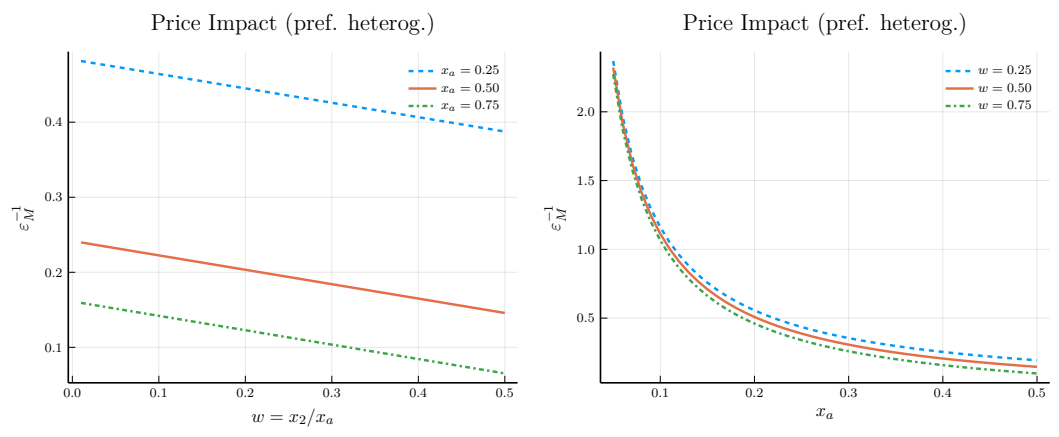
where $\mathbb{E}^u\left[\gamma_j\right] \equiv (1 - w)\gamma_1 + w\gamma_2$.

Determinants of the market elasticity: Preference heterog.

$$\varepsilon_M^{-1} = \left(1 - \psi^{-1}\right) \frac{\gamma\sigma^2}{y_0(X)} \left[\frac{1 - \overline{\alpha}_p}{x_a} - \frac{\gamma_0 - \mathbb{E}^u\left[\gamma_j\right]}{\gamma}\right] + \mathcal{O}\left(\epsilon^2\right).$$

- With heterogeneity, agg. elasticity is finite even when $\overline{\alpha}_p = 1$
- Heterogeneity attenuates the price impact if $\gamma_0 \geq \gamma_1 \geq \gamma_2$.
  - more elastic markets due to risk misallocation
  - passive buys from active investors, depressing prices
- State dependence beyond active wealth share:
  - wealth distribution **among active investors** , w, also matters ($\mathbb{E}^u\left[\gamma_j\right]$)
  - reminiscent of models with heterog. intermediaries (Silva, 2020; Kargar, 2021)

Inverse market elasticity: Preference heterogeneity



Portfolio constraints

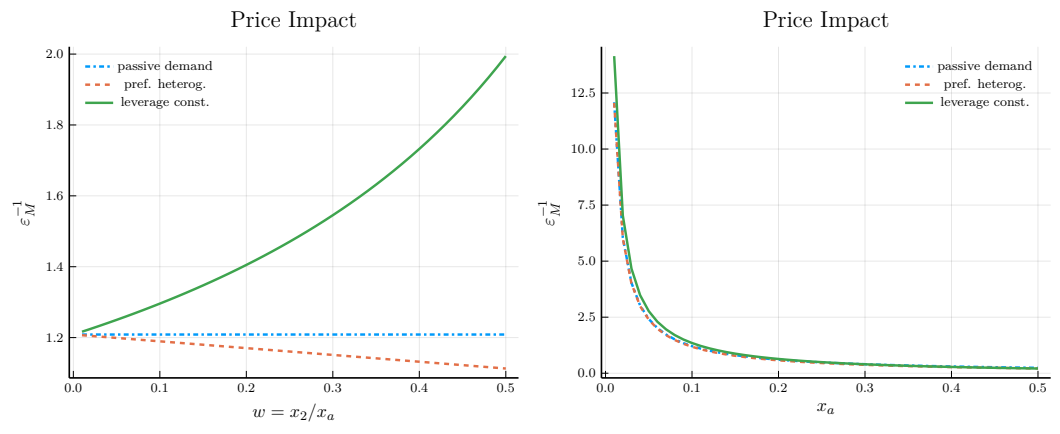
Proposition (Aggregate elasticity: portfolio constraints)

When active investors have heterogeneous preferences and also face leverage constraints, the inverse aggregate market elasticity, $1/\varepsilon_M$, is given by:

$$\varepsilon_M^{-1} = \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{y_0(X)}\left[\frac{1-(1-x_c)\overline{\alpha}_p - x_c\frac{\overline{\sigma}}{\sigma}}{x_u} - \frac{\gamma_0 - \mathbb{E}^u\left[\gamma_j\right]}{\gamma}\right]$$

$$+\underbrace{\left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{y_0(X)}\left[\frac{\mathbb{E}^u\left[\gamma_k\right]-\mathbb{E}^c\left[\gamma_k\right]}{\gamma x_u} + \frac{\gamma\sigma\left(1-\psi^{-1}\right)}{2y_0(X)}\frac{\gamma_0-\mathbb{E}^u\left[\gamma_k\right]}{\gamma}\right]x_c}_{\geq 0} + \mathcal{O}\left(\epsilon^2\right)$$

Inverse market elasticity: impact of different frictions



- Leverage constraints make the aggregate market more inelastic

Decomposition of the Aggregate Market Elasticity

Decomposing the market elasticity

- Inverse elasticity

$$\varepsilon_M^{-1} = \frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1}{y_0(X)} \frac{\partial y_2(X)}{\partial F} \epsilon^2 + \mathcal{O}\left(\epsilon^2\right)$$

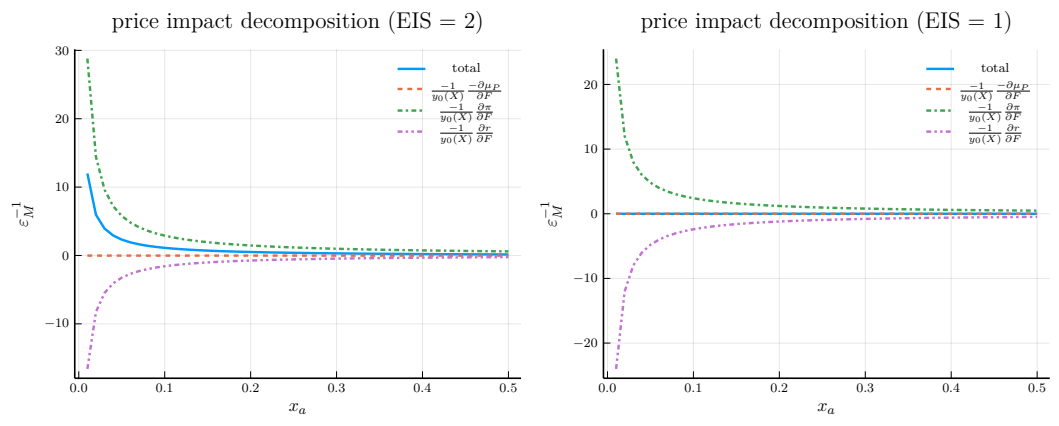
- Pricing condition

$$y_t = r_t + \pi_t - \underbrace{\left(\mu - \mu_{y,t} - \sigma_{y,t}\sigma_{R,t}\right)}_{\mu_{P,t}}$$

- Can decompose the aggregate elasticity into three terms:

$$\varepsilon_M^{-1} = -\frac{1}{y_0(X)} \left[\frac{\partial r_2(X)}{\partial F} + \frac{\partial \pi_2(X)}{\partial F} - \frac{\partial \mu_{P,2}(X)}{\partial F} \right] \epsilon^2 + \mathcal{O}\left(\epsilon^2\right)$$

Inverse market elasticity: decomposition



- r and π move aggregate elasticity in opposite directions

Next steps

- Bring back the flow shocks and quantitatively evaluate its importance
- Calibrate the model to the Flow of Funds with multiple sectors
  - rich heterogeneity in the model is key to match the data
  - challenging: solving and estimating a high-dim. macro-finance model
- Can frictions generate observed levels of market volatility?
  - flow volatility
  - high price impact?

Beyond Perturbation

Pitfalls of the perturbation method

- Analytical results allows for deeper insights of the model
- Quantitative results?
  - Increasing the perturbation order doesn't necessarily improve the approximation's accuracy.
  - Diverges for *large* perturbations

Beyond perturbation: The Homotopy Analysis Method (HAM)

- Builds on the perturbation techniques
- Allows for analytical results
- Provides mechanisms to ensure convergence of the Taylor expansion of the value function
- More suited for quantitative work

Fundamental idea of state-global perturbation

1. Find a "similar" problem where drifts and vols are constant:
2. Partial differential equations become algebraic (linear) equations

- Generic HJB:

$$U - \delta V + \mathbb{E}\left[\frac{dV}{dt}\right] = 0$$

- We are looking for a Taylor expansion of V:

$$V(s) = V_0(s) + \epsilon V_1(s) + \epsilon^2 V_2(s) + \dots$$

- Because drifts and vols are 0 for the reference model, we can write:

$$V_k = \frac{U_k + \mathbb{E}\left[\frac{dV}{dt}\right]_k}{\delta}$$

Illustration: The Two Trees model: Model setup

- Two Trees (Cochrane, Longstaff, and Santa-Clara, 2008)
  - Consumption is sum of the dividends of two trees
  - The dividend process for each tree is given by GBM
  - The correlation between the Brownian Motions is ρ
  - Representative agent with CRRA utility
  - Relevant state variable:

$$s = \frac{D_0}{D_0 + D_1}$$

  - Defining $V \equiv pd \cdot s$, it can be show that V satisfies:

$$s - \delta V + \mathbb{E}\left[\frac{dV}{dt}\right] = 0$$

Illustration: The Two Trees model: Exact solution

figures/2trees_closed_form.png

Illustration: The Two Trees model, a reference model

- If dividends are perfectly correlated, we have a Lucas economy, with constant price-dividend ratio.
- Can parameterize the correlation btw. the two trees as: $\rho_\epsilon(\epsilon) = 1 + \epsilon(\rho - 1)$

figures/2trees_0th_order.png

Illustration: The Two Trees model, 1st order perturbation

figures/2trees_1st_order.png

Illustration: The Two Trees model, 2nd order perturbation

figures/2trees_2nd_order.png

The Homotopy Analysis Method (1): Basic formulation

- Deformation equation

$$(1-\epsilon)(V-V_0) + h \cdot \epsilon \cdot HJB(V,\epsilon) = 0$$

- For $\epsilon = 0$, the deformation equation yields: $V = V_0$
- For $\epsilon = 1$, the deformation equation yields: $HJB(V, \epsilon = 1) = 0$

The Homotopy Analysis Method (2): More flexible formulation

- Deformation equation

$$(1-\epsilon)(V-V_0) + h \cdot \alpha(\epsilon) \cdot HJB(V,\epsilon) = 0$$

- For $\epsilon = 0$, the deformation equation yields: $V = V_0$
- For $\epsilon = 1$, the deformation equation yields: $HJB(V, \epsilon = 1) = 0$

The Homotopy Analysis Method (3): More flexible formulation

- Deformation equation

$$(1-\epsilon)(V-V_0) + h \cdot \alpha(\epsilon) \cdot HJB(V,\epsilon) = 0$$

- Still a linear algebraic equation!
- Since we are working with Taylor expansions, choosing the function α is equivalent to choosing a finite vector of with the same shape as the perturbation order.

$$\alpha(\epsilon) = H_1\epsilon + H_2\epsilon^2 + \ldots + H_k\epsilon^k$$

- $h, H_1, H_2, \ldots$ are (hyper)parameters that we can tune.

Two Trees model with HAM

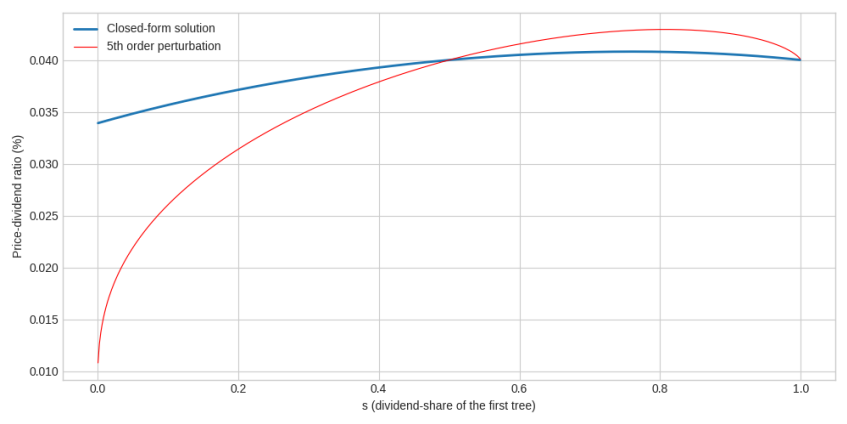
- Link to code

Two Trees example: HAM with meta-optimization

- Choose a metric to evaluate the solution
- We will use mean squared error of the residuals of the HJB

$$\text{Performance}(\text{hyperparameters}) = MSE\left(\text{solution}\left(\text{hyperparameters}\right)\right)$$

Two Trees example: HAM with meta-optimization (2)



link

HAM - Practical issues

- Recursive formulation is not practical because of nested derivatives

$$V_k = \frac{U_k + \mathbb{E}\left[\frac{dV}{dt}\right]_k}{\delta}$$

- Solution: project the value function using a basis (Neural Tangent Kernel) and do **one high-order** Taylor expansion

- How implement high-order Taylor expansions?
  - Taylor-Mode Automatic Differentiation for Higher-Order Derivatives in JAX (2019)

HAM with Kernels

- Impose:

$$V(s) = V_0(s) + \epsilon\phi(s)^T\Theta_1 + \epsilon^2\phi(s)^T\Theta_2 + \dots$$

- The problem is still linear! Now in Θ.
- Each term Θ_k can be solved with linear least squares
  - $\Phi\Theta_k = \mathit{target}(\Theta_0, \Theta_1, ..., \Theta_{k-1})$
  - Because the same Φ shows up for every order, just need one SVD/Cholesky decomposition.
- 5th order solution run time for the 2 Trees example: 5ms

Next steps

- Use this technique to solve our model
- Allows for quantitative results

Thanks!

Second-order terms

Back

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this ext page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.